Visualising Emotion

Does an AI drawing carry the feeling in the words that made it?

Shre Hari Prem Anandh
MSc Artificial Intelligence · University of Sheffield
Supervisor: Dr Chaona Chen

1,025 real self-reports · 1,012 analysed · 28/28 tests passing

THE PROBLEM

Some prompts can be checked. Feelings cannot.

“a red car”

- Is there a car?
- Is it red?
- Done — the prompt names objects.

This is what almost all text-to-image evaluation does.

“tired but happy”

- No object to look for.
- No answer key.
- And the feeling is mixed.

Nobody had built a way to check this.

Median prompt in this dataset: four words. Mixed feelings are common.

THE DATA

A dataset that cannot be downloaded

Festival of the Mind 2024, Sheffield. Members of the public typed how they felt; DALL·E drew each entry as black-and-white line art.

1,025
drawings generated
0 processing failures

1,012
analysed
(13 were pre-festival tests)

607
prompts carry a feeling
405 do not — the control

87
prompts drawn more than once
covering 336 records

Three traps before any analysis was possible

- Three incompatible filename schemes in one folder — a naive loader resolves under 20% of the corpus.
- Hidden C2PA timestamps inside every PNG revealed that records 1–13 were made before the festival.
- 87 prompts were drawn more than once — which looked like a nuisance and became a finding.

THE GAP

Four literatures, and the hole between them

Emotion in text	Well studied. GoEmotions gives 27 categories and a classifier — but short first-person self-report is the hardest case.
Emotional image generation	Well studied. And the strongest evidence says colour, not shape, carries most emotional meaning.
Text-to-image evaluation	Well studied — for objects. The metrics check whether the picture contains what the prompt named.
Reading emotion from an image	Possible but weak. Vision-language models are positively biased, order-sensitive readers.

Nobody had asked whether the feeling in a person's own unedited words survives the journey into a drawing.

97 papers collected and read · 73 kept as core evidence · organised into twelve themes

THE RESEARCH QUESTION

To what extent does an automatic image-emotion reader agree with prompt-derived emotion labels for AI-generated monochrome drawings — and how does that agreement vary by visual content and by repeated prompts?

This measures agreement between two automatic systems. It does not establish that a human viewer perceives the same feeling — and that distinction is carried honestly through every claim I make.

METHOD

Describe first. Never ask for the emotion.

01 Describe

Five plain factual questions to the image model: expression, action, posture, how many people, what is in the picture.

02 Read

A separate language model infers a feeling from that description. A third model reads the person's own words.

03 Compare

Four measures of closeness — every one referenced to chance.

The vision model is never asked “what emotion is this?” — that is the supervisor's design, and it is measurably better.

METHOD · STAGE ONE

The five questions asked of every drawing

1. What is the facial expression?
2. What is the person doing?
3. What is the body posture?
4. How many people are there?
5. What is in the picture?

Then, one neutral sentence

“A drawing of one person showing a park. The person is sitting, hunched. The facial expression is sad.”

A second model reads a feeling out of that sentence — never out of the picture.

Honest caveat: the expression answer is often already a feeling word, so the two stages are not fully independent. This is reported.

METHOD DEVELOPMENT

Four things that failed first

✕	**GoEmotions on the descriptions**	Collapsed to “neutral” above 95%. It needs first-person feeling words; a visual description has none.
✕	**A single image caption**	Too terse — “a man with hands in the air”. The reader collapsed to “surprise” for almost everything.
✕	**Qwen2.5-VL, 3B parameters**	Exceeded a 16 GB laptop. Thrashed into swap: 4 hours 45 minutes, zero progress.
✕	**Asking CLIP for the emotion**	Beaten head-to-head — and inspection found arms-raised celebration read as anger, 0.73.
✓	**BLIP-VQA**	0.4B parameters, ~1.5 s per drawing — and its “facial expression” answer is exactly the cue the next stage needs.

RIGOUR

Every number is measured against luck

Within each subset, the drawings are shuffled and re-paired against the untouched prompts, and all four measures are recomputed. Repeated 300 times, with a fixed seed.

Real pairing



each drawing with its own words

Shuffled — 300 times



each drawing with a stranger's words

A result counts only insofar as it beats the shuffle. Every figure in this talk is quoted against its own null.

RIGOUR · THE VALIDITY CHECK

A control that passes

405 of the 1,012 prompts contain no feeling at all.

“pigs”

“mickey mouse”

“disney characters”

If my method were inventing emotion, it would score above chance here.

76.6%	measured	76.6%	pure chance	z = +0.3

It finds emotion only where emotion exists. That is why the other numbers can be believed.

RIGOUR · CALIBRATING THE SCALE

Chance is not zero

The similarity measure cannot reach 0 — and randomly paired items already score 76.

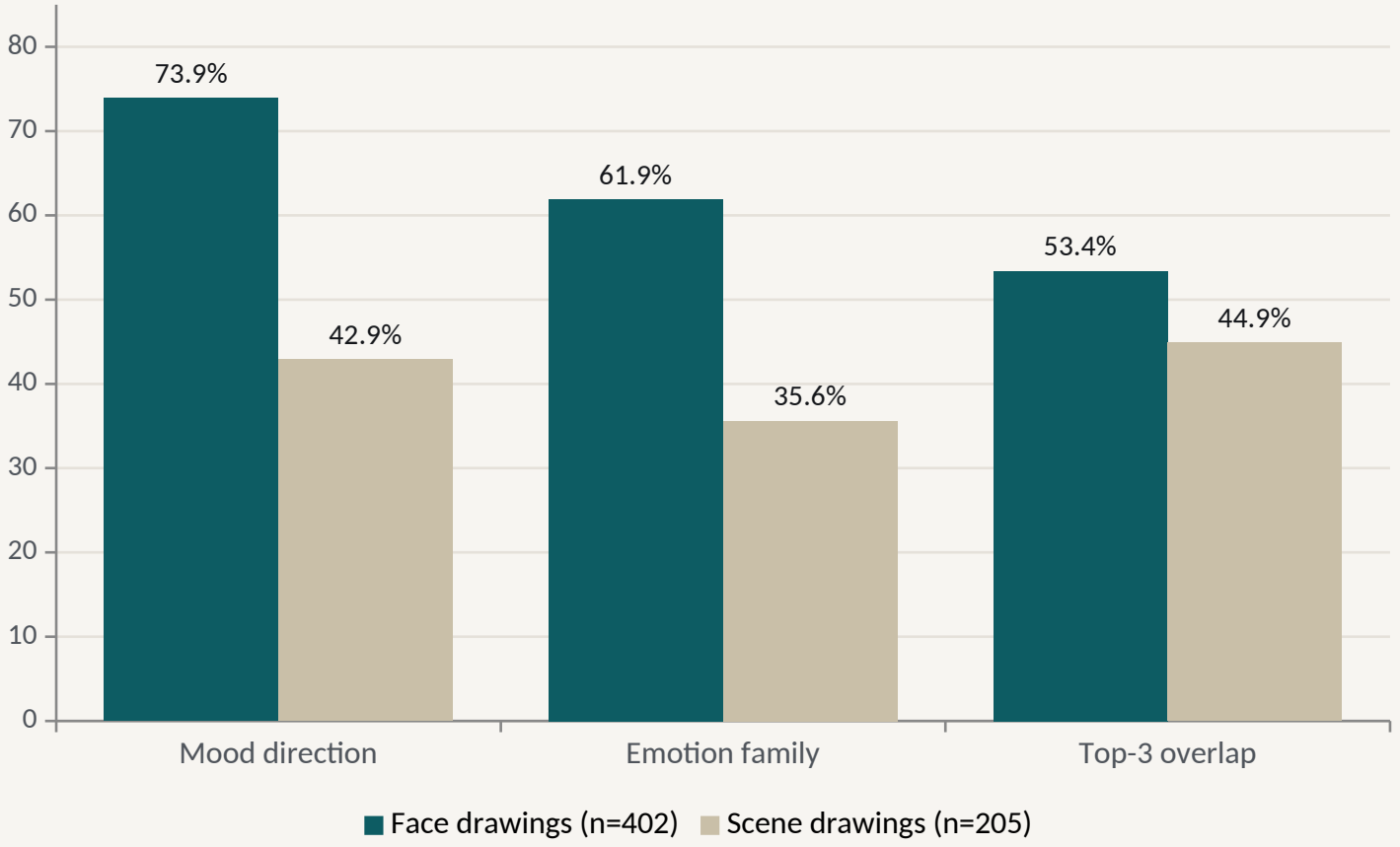


**A score of 71 looks like a pass.
It is below chance.**

So I changed what I report: mood-direction and emotion-family agreement now lead, and this score is always shown against its chance line.

FINDING ONE

Alignment is far stronger when there is a face



1.7×

better alignment on both mood direction and emotion family

Stated carefully

The pipeline finds stronger alignment for drawings it classifies as containing a readable face. That classification and the emotion reading share a source — so this is partly circular, and I say so.

All above chance (z +4.4 to +10.4). Holds on distinct prompts only: face 71.8% vs scene 37.3%.

FINDING TWO

The generator is inconsistent with itself

87 prompts were drawn between 2 and 42 times each. Same words in — different feeling out.

68.8%

of the time the redrawn image conveys the SAME emotion

28 / 87

repeated prompts are fully consistent across all their drawings

336

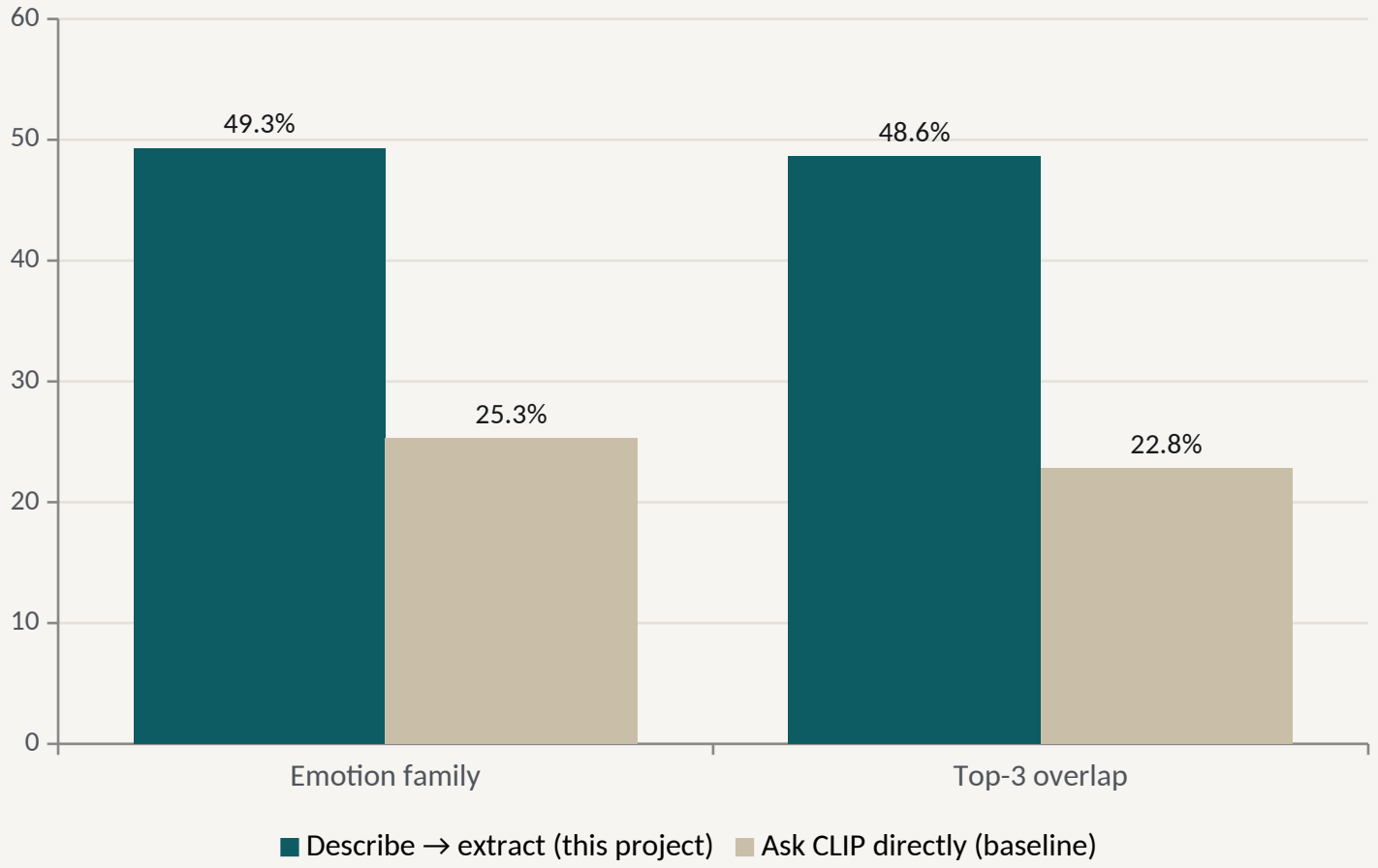
records sit inside a repeated-prompt group

A single drawing is one sample of the model's behaviour — not a fixed answer.

That is a caution for anyone evaluating a generative model on a single output — including the standard benchmarks.

FINDING THREE

The architecture choice was worth 24 points



Head-to-head on the same records, with the prompt side held constant and only the image reader changed.

Roughly double

So the supervisor's redirection was empirically justified, not simply accepted. Asking a vision model for an emotion directly performs poorly on line art.

83.7% vs 74.2%

on valence–arousal similarity — the measure that cannot separate faces from scenes

BUILT ALONGSIDE

A live instrument, not a screenshot

The pipeline runs in a browser: pick a drawing or upload one, watch the five answers appear, see both readings side by side, and see the similarity scored against chance.

Choose a drawing

from the dataset, or upload your own image

Watch it described

five factual answers — emotion never asked

Compare

both readings, and the score against chance

Every number it shows is referenced to chance

- the axis starts at the measure's real floor of 41, not at 0
- the middle half of random pairings is bracketed on the scale
- the headline reads “closer than N% of random pairings”
- nothing typed or uploaded is stored, and no visitor IP is logged

DATA PROTECTION

Redacting names without breaking the data

Supervisor's direction: redact real names, keep illness and grief content — it is not identifying on its own and it is essential to the analysis — and document the procedure.

1 Detect

77% of the corpus is lower case, so capitals alone would miss most names. A 7,944-name gazetteer, a 236,000-word dictionary, and naming phrases.

2 Review

Every candidate carries a recorded decision and a reason. The audit fails if anything is unreviewed.

3 Apply

Scoped to the specific record each name was reviewed in.

Two errors the audit caught

- A bare name list destroyed “Jack Frost” and “woolly mammoth”.
- Redaction forged a duplicate: two different names both became [name].

Redaction is analytically inert

Re-running the text model on the redacted prompts moves each probability by at most 0.012 and changes zero emotion readings. 8 records held names; all 1,012 are still shown.

HONESTY

What I cannot claim

This is automated agreement, not human perception	Both readers are models and they disagree in opposite directions: CLIP skews negative on line art, the description model reads 57% of readable faces as happy.
The face / scene split is partly circular	It is defined by whether the model reported an expression — and the emotion is read from that same description.
The valence–arousal score is a designed proxy	Built on hand-set emotion coordinates. Not a continuous lexicon rating, and it cannot separate faces from scenes.
No claim about colour, and none about direction	The corpus is monochrome but there is no colour condition. An earlier positive-versus-negative claim was withdrawn.

A human-rating study was designed in full and deliberately not run — approval alone would have taken a fortnight. It is the clear next step.

IN SUMMARY

What this contributes

Not a novel emotion classifier.

- **A question nobody had asked**

  of real, first-person, mixed self-report

- **A dataset that cannot be downloaded**

  1,025 real emotional entries with their drawings

- **An evaluation design that holds up**

  null baselines throughout, and a control that passes

- **A measured method comparison**

  the architecture was justified, not assumed

- **A generative-consistency finding**

  the same prompt does not convey the same feeling

- **A documented data-protection procedure**

  tested, and proven not to change any result

Thank you — questions welcome

Next step: human validation of the automatic readers